

Algebra 1
Unit 9
Lesson 2 Homework

Name Answer key
Date _____
Period _____

Use the information provided in each graph to write the equations in vertex form and standard form.

1. Identify the points.

a. Vertex: $(-1, -4)$

b. Point: $(1, -3)$

2. Write the equation for this quadratic in vertex form and standard form.

a. Vertex form:

$$y = \frac{1}{4}(x + 1)^2 - 4$$

$$\begin{aligned} -3 &= a(1 + 1)^2 - 4 \\ -3 &= 4a - 4 \\ 1 &= 4a \\ \frac{1}{4} &= a \end{aligned}$$

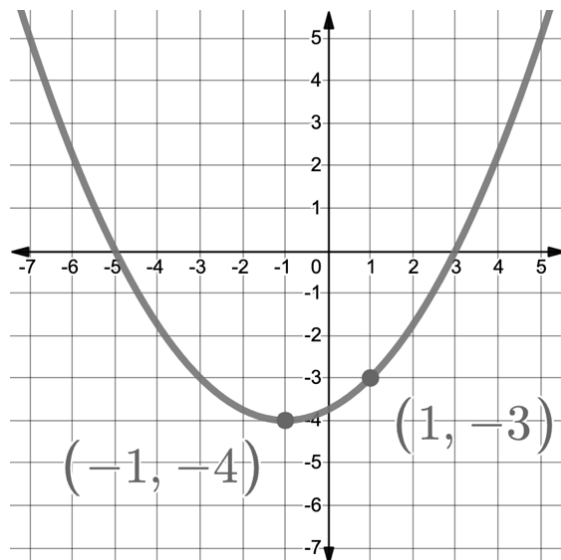
b. Standard form:

$$\frac{1}{4}(x + 1)(x + 1) - 4$$

$$\left(\frac{1}{4}x + \frac{1}{4}\right)(x + 1) - 4$$

$$\frac{1}{4}x^2 + \frac{1}{4}x + \frac{1}{4}x + \frac{1}{4} - 4$$

$$y = \frac{1}{4}x^2 + \frac{1}{2}x - 3.75$$



3. Identify the points.

a. Vertex: $(-1, 4)$

b. Point: $(-2, 2)$

4. Write the equation for this quadratic in vertex form and standard form.

a. Vertex form:

$$y = -2(x + 1)^2 + 4$$

$$\begin{aligned} 2 &= a(-2 + 1)^2 + 4 \\ 2 &= a + 4 \\ -2 &= a \end{aligned}$$

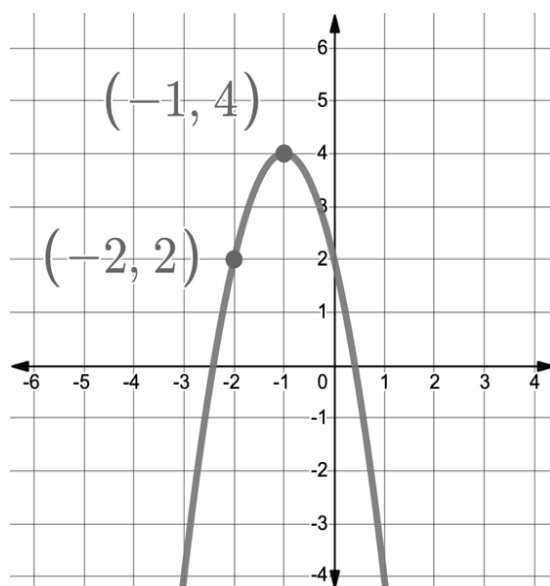
b. Standard form:

$$-2(x + 1)(x + 1) + 4$$

$$(-2x - 2)(x + 1) + 4$$

$$-2x^2 - 2x - 2x - 2 + 4$$

$$y = -2x^2 - 4x + 2$$



5. Identify the points.

a. Vertex: (1, -4)

b. Point: (3, 0)

6. Write the equation for this quadratic.

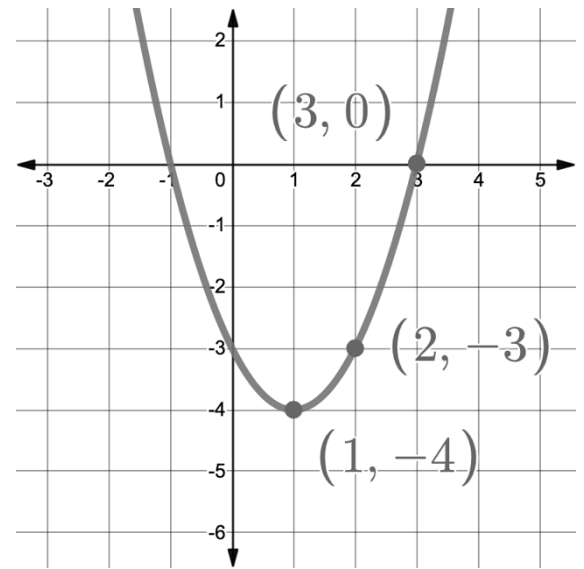
a. Vertex form:

$$y = (x - 1)^2 - 4$$

$$\begin{aligned} 0 &= a(3 - 1)^2 - 4 \\ 0 &= 4a - 4 \\ 4 &= 4a \\ 1 &= a \end{aligned}$$

b. Standard form:

$$\begin{aligned} (x - 1)(x - 1) - 4 \\ x^2 - 1x - 1x + 1 - 4 \\ y = x^2 - 2x - 3 \end{aligned}$$



7. Identify the points.

a. Vertex: (2, 1)

b. Point: (4, 0)

8. Write the equation for this quadratic.

a. Vertex form:

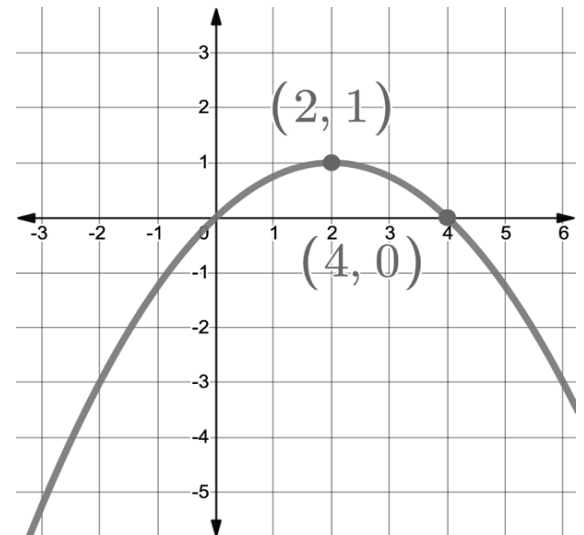
$$y = -\frac{1}{4}(x - 2)^2 + 1$$

$$\begin{aligned} 0 &= a(4 - 2)^2 - 1 \\ 0 &= 4a + 1 \\ -1 &= 4a \\ -\frac{1}{4} &= a \end{aligned}$$

b. Standard form:

$$y = -\frac{1}{4}(x - 2)(x - 2) + 1$$

$$\left(-\frac{1}{4}x + \frac{1}{2}\right)(x - 2) + 1 \Rightarrow -\frac{1}{4}x^2 + \frac{1}{2}x + \frac{1}{2}x - 1 + 1 \Rightarrow y = -\frac{1}{4}x^2 + x$$



9. Identify the points.

a. Vertex: (0, -6)

b. Point: (1, -3)

10. Write the equation for this quadratic.

a. Vertex form:

$$y = 3(x - 0)^2 - 6$$

$$\begin{aligned} -3 &= a(1 - 0)^2 - 6 \\ -3 &= a - 6 \\ 3 &= a \end{aligned}$$

b. Standard form:

$$\begin{aligned} 3(x)(x) - 6 \\ y = 3x^2 - 6 \end{aligned}$$

